Terraform Plan vs Apply: Beginner's Deep Dive

Understanding what happens behind the scenes during Terraform execution prevents costly production downtime. This guide breaks down how Terraform reads code, checks real infrastructure, and safely updates your cloud environment.

What you'll learn:

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1. Reading the Terraform Plan Output

Running terraform plan acts as a dry run that compares three things: your code, state file, and real cloud environment. It identifies missing or modified resources and assigns symbols: + means creation, - means deletion, and ~ means update in-place. Reviewing this output helps you catch accidental resource replacements before they happen in live environments.

```
$terraform plan
```

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2. Locking Execution Plans with Output Files

In team setups, real cloud resources can change between running plan and running apply. Saving your plan file guarantees that Terraform executes the exact actions you reviewed. It captures the exact deployment state into a binary file that apply can read directly without recalculating changes.

```
$ terraform plan -out=tfplan.binary
  terraform apply tfplan.binary
```

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3. What Happens During Terraform Apply

When you run apply, Terraform builds a dependency tree of your infrastructure components. It determines the order of operations, like making sure virtual networks exist before placing servers inside them. Once calculated, it sends HTTP requests to your cloud provider's API and updates your state file with new resource IDs.

```
$ terraform apply
```

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4. Modifying Single Components with Target Flags

Sometimes you need to deploy a quick bug fix without risking updates to the rest of your architecture. The target flag forces Terraform to isolate a specific resource block during execution. This limits both plan calculation and apply actions strictly to the specified resource name.

```
$ terraform apply -target=aws_s3_bucket.my_app_bucket
```

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Key Takeaways

- Always review the execution output of terraform plan line by line before applying.
- Store state files in remote storage like AWS S3 with state locking enabled.
- Save plan outputs to binary files when running Terraform inside automated CI/CD pipelines.
- Run terraform fmt regularly to keep your code clean and readable for your team.
- Use targeted runs only for emergency fixes, then return to full stack plans.

Watch Out For

- Running terraform apply directly with the auto-approve flag without reviewing plans.
- Manually modifying the local terraform.tfstate file in a text editor.
- Hardcoding API keys or secret credentials directly into your Terraform configuration files.
- Making manual changes in the AWS or Azure web console while using Terraform.

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